

Mixed-Methods Self-Tracking: Qualitative Styles of Quantifying the Self

Hannah R. Nolasco
df104001@st.osakafu-u.ac.jp
Osaka Prefecture University
Osaka, Japan

Andrew W. Vargo
awv@omu.ac.jp
Osaka Metropolitan University
Osaka, Japan

Koichi Kise
kise@omu.ac.jp
Osaka Metropolitan University
Osaka, Japan

ABSTRACT

Self-tracking technologies face inconsistent reactions from users that range from excitement and enthusiasm to frustration and discomfort. Many of those on the latter end of this spectrum deal with complex feelings of disempowerment and objectification; they also face adverse effects from the excessive quantification of their lived experiences. Rather than serve as active agents in their process of monitoring and improving themselves, users feel reduced to docile spectators who are coerced into changing their behavior. This position paper explores the tensions in power redistribution and depersonalization present in existing self-tracking methods and contemplates on qualitative alternatives that may improve the design of future technologies, particularly in the field of learning.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; **Ubiquitous and mobile computing design and evaluation methods**.

KEYWORDS

self-tracking, datafication, disempowerment, objectification

ACM Reference Format:

Hannah R. Nolasco, Andrew W. Vargo, and Koichi Kise. 2022. Mixed-Methods Self-Tracking: Qualitative Styles of Quantifying the Self. In *Proceedings of the 2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC '22 Adjunct)*, September 11–15, 2022, Cambridge, United Kingdom. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3544793.3560386>

1 INTRODUCTION

One may summarize the function of a self-tracking technology's quantification as an implicit or explicit medium for influencing user behavior [2], but the opposite can also be true: By negotiating your priorities with your device, you can define for yourself what behaviors you wish to encourage or discourage. This could be seen as one of the many forms of empowerment conferred to you by your Quantified Self (QS), as the visible measurements collected by your chosen technology afford you the knowledge and insight

to make informed decisions about your health and wellbeing [5]. Self-tracking thus becomes a way for you to assert control over your life [6, 17, 21, 24] and reinvent yourself through rigorous self-examination [10]. Another manner to interpret this, however, would be by perceiving the connection between yourself and your application as a power dynamic. However way you outline your goals to a device, the technology gains the power to impose activities that are meant to drive you towards a stipulated image of your most ideal self whose statistics are not determined by you—you don't control the tool, you are simply choosing how the tool controls you [8]. There is an element of anthropomorphism present in this evaluation as users wind up viewing their interaction with their device as a real relationship: They describe their adjusted behaviors as having been "forced" upon or "demanded" of them, and should they fall short in appeasing the machine's algorithms, they experience feelings of guilt and the compulsion to justify any poor results [8]. In order to avoid disappointing or upsetting themselves, users will adapt their behavior and adjust the deadlines of their objectives to ensure that they receive positive feedback from their device [8]. When these things happen, wearables become fetters instead of augmentations to daily living, constantly calling attention to uncomfortable truths about one's physical body to the point of frustration and annoyance [4, 12, 19, 24]. This impacts the likelihood that one will adhere to a self-tracking tool: They may even opt to forego usage entirely [8]. While there are legitimate merits to the use of such technologies, there is no definitive way to prove it holds no deleterious effects; the often mixed reception it receives has yet to be wholly intervened with. It may be necessary for researchers to audit their work more thoroughly and consider how well self-tracking accommodates the nebulous quality of self-improvement and efficiency. This position paper invites discourse over the idea that incorporating more qualitative measurements can improve how people adhere to and interact with self-tracking technologies: We contemplate on the power that are we handing to these tools when determining how we should function and operate as human beings, as well as how the ramifications of such a power redistribution influences the effectiveness and reach of these kinds of devices. We then consider the ways in which unconventional and qualitative approaches to self-tracking can be applied to future studies relating to improving learning outcomes.

2 PROBLEM STATEMENT

Tracking technology can serve a purpose outside of surveillance and control and still be transformed into a means of subjugation. Rather than be used as a reference for adjusting training intensity for the health and safety of football athletes for example, a GPS device can instead be used to measure accomplishment and sanction the

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
UbiComp/ISWC '22 Adjunct, September 11–15, 2022, Cambridge, United Kingdom
© 2022 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-1-4503-9423-9/22/09...\$15.00
<https://doi.org/10.1145/3544793.3560386>

underperforming [14]. Being able to examine yourself in broader strokes rather than granular details can at times be the healthier way to develop and grow, and it prevents you from perceiving your data as “someone” who should know you better than you know yourself [28]. The question now is: What aspects of personhood does self-tracking diminish in its process of quantifying the self? There is a risk that we are reducing individuals to numbers [27] and diminishing the self-monitoring of health, wellness, and productivity into simple linear processes—how do we reintroduce a sense of humanity into the act of self-tracking to ensure that it appeals to a broader audience without asking them to relinquish control over their lives?

3 DISCUSSION AND RELATED LITERATURE

This section delves into the mixed reception faced by highly quantitative self-tracking methods. There are elements of **disempowerment**, **objectification**, and **power redistribution** persistent in the act of quantifying the self: These factors are important to acknowledge and consider in the process of identifying where sensing and tracking can be improved further—potentially through more qualitative means.

3.1 Ambiguous Empowerment

There are tensions evident in the polarizing experiences that users have with self-tracking [13]. Many perceive the practice as playful and exciting due to the increased self-awareness that it provides [23], which in turn gives one the agency to reinvent oneself based on the personal knowledge accrued [10]. Others also see it as a form of emancipation or a means of rebelling against society’s strict and rigid conventions—by taking power away from a system of authority (i.e. measuring one’s weight independently instead of having a medical physician do it), they are able to cultivate and maintain a sense of greater autonomy [25]. Indeed, it can be liberating to believe that your successes and failures are solely your responsibility and that you alone are their cause [20]. For those on the opposite end of the spectrum however, the inverse is their reality: The data feels reductionist and inaccurate, incapable of thoroughly capturing the full extent of one’s condition and therefore is unreliable [16]. Despite this sentiment, there are others who find themselves building their entire lives around their data, changing many of their minute behaviors for the sake of appeasing their chosen tracking technology, despite the fact that doing so brings them frustration and discomfort [16]. Suddenly they are recalibrating the way they do everyday activities that should have once been second nature, such as walking, exercising, eating, and sleeping [8]. It becomes necessary to “negotiate” with their device in order to find a habit that suits both algorithm and user [25], otherwise they relegate their decision-making power to the machine [8]. This is done with the assumption that their device knows better and holds the authority over what is “good” and “healthy” [1, 20, 25, 28]. For some it is a positive experience: A viable way to simplify the complexities of life that they as an individual can survive without (e.g. meal planning). For others, being given concrete categories is limiting and depersonalizing [25]. It is for this reason that there are several texts that criticize the liberation and agency that some dedicated self-trackers espouse; the algorithms behind these self-tracking

systems still inevitably promote a normative way of living, a right method and a wrong method, which it reinforces through numbers and scores [20, 25] as well as astute forms of persuasion [7]. No matter how neutral we perceive numbers to be, they can still dictate and perpetuate biases [20, 25]. Furthermore, when we focus exclusively on the aspects of life that we deem categorically quantifiable (e.g. sleep quantity, calories consumed and burned, protein and fat intake distribution), we neglect to consider the more subjective yet equally meaningful phenomena that are essential to completing the picture of a real life human being. It becomes even more problematic when we reduce rich experiences such as emotions and productivity into mere numerical information that can be rigged for greater efficiency [25].

3.2 The “Someone” becomes “Something”

There are compelling arguments critical of the QS movement that assert that self-tracking reduces people to numbers [27]. Rather than explore the subjective depth of one’s own personal lived experiences, people choose to refer to themselves by their statistics (e.g. calories eaten, calories burned, weight gained or lost) [8]. They see themselves through the lens of the technology’s observations and lose the ability to trust in their own self-assessments [25, 28]. In order to conform to the normalization of the “ideal self,” they wind up perceiving themselves as objects [9] that must measure up in order to be considered up to standard. In short, people are reduced to an instrument that merely needs fine-tuning—they no longer play an active role in their improvement and instead become spectators of their device’s automated calculations [25, 28]. It does not help that the marketing of self-tracking can often be moralistic and judgmental [25]. Because numbers must be accurate and precise in comparison to our own natural instincts, then it must be better. Choosing not to self-track starts to seem like a moral failing on one’s part as you are not maximizing the process of achieving your best self [25]. The sentiments of the QS movement seem to declare that if you could use hard evidence for self-improvement rather than trial and error, then you should want to [11]. Such an assumption does not acknowledge that there are people who build more sustainable and sincere habits through the natural, non-linear process of bettering oneself—self-tracking can sometimes make the technology the reason for one’s shift in lifestyle (i.e. they feel obliged to appease their tracking device), rather than the real enriching benefits that it can provide. It begs the question: “is this model pushing me closer in contact or further away from myself and my world?” [28]. Bodies are heterogeneous [8], and a person’s unchangeable characteristics (e.g. the presence of a disability) and socioeconomic conditions can impact their ability to do the things most tracking systems summarize to suit only limited contexts [15]. Failing to consider the nuanced experiences faced by users can compromise the accuracy of tracking technology [8].

3.3 Perennial Power Structures

Another danger present in self-tracking is that we may only be omitting responsibility where it is due: Rather than holding society accountable for its strained conditions (e.g. harsh working environments, unrealistic body expectations, systemic inequalities, expensive healthcare), we are omitting its responsibility to cater

to our needs in favor of allaying all the pressure of achieving wellness on the individual [20]. Power is handed to the machine to define what “healthy” should mean to us, and the constant micro-surveillance—whether real or imagined—coerces us to obey and behave in a certain way [8]. Even when the technology does not constantly intervene in the user’s life, its regular notifications work to maintain its authority [8]. When the source of the knowledge gathered and reported is unclear or unknown, it reinforces the device’s power further [8].

An added dimension emerges when companies, corporations, and even medical institutions oblige people to self-track as a way to monitor themselves on the establishment’s behalf: Those who self-monitor as instructed by their place of work feel distrusted and will err on the side of using self-tracking technologies superficially [26]. Self-tracking of their productivity at work may also become tied to their health insurance [22] and opportunities for promotion or salary increase [18], making self-tracking an imposed activity rather than a voluntary one [20]. People are also fatigued, stressed, and unhappy for reasons that are outside of their control, and not even a statistically-supported lifestyle can address such factors. In the realm of self-tracking and sensing, it may be pertinent to consider that empowerment may not simply be the act of taking or negotiating control but the decision to live without any—empowerment could also mean not changing, remaining the same, or even disengaging from examining oneself with so much scrutiny. It could also be the decision to lead a flawed, imperfect life where “efficiency” is given its own unique definition outside of the guidance of an algorithm or a status quo.

There are those who appreciate self-tracking outside of the perceived zero-sum game that the QS movement is often criticized over—instead of treating the numbers as objective truth, certain individuals track for the virtue of engaging the world around them. In an article by Sharon and Zandbergen [25] that covers alternative ways of approaching self-tracking, several users stated that their ultimate goal was not to employ the numerical data for the sake of self-improvement but to gain an additional and augmented “sense” of themselves or their surroundings. One user stated that they took a picture at 08.36 every single day to gain a more acute awareness of whenever it was that time of day. Other users used self-tracking to develop an increased consciousness for their habits and patterns. By cultivating these extra senses, these self-trackers believe that they will eventually no longer have a need for the tools being used to track and can intuit things themselves. Another user from the same study, Alberto, would take photos of objects that he would hold in his right hand, simply as a way to build a “DNA code” of his life. In his situation, the “act of recording” was what made self-tracking meaningful to him. It is possible that the act of monitoring oneself could also be used in contexts outside of self-improvement, and engaging in such divergent approaches may soften the hard edges that make self-tracking unattractive for certain individuals.

4 FUTURE WORK

The previous section outlined the negative effects and untoward symptoms of highly quantified and analytical self-tracking. When people are reduced to numbers and made to objectify themselves in order to conform to certain norms, it leads to data fetishism. Users

become docile or frustrated in the face of the power they are asked to relinquish by letting their tracking device have authority over their decisions and perceptions of health and self-improvement. This power struggle is exacerbated by the moralistic and judgmental way that sensing technology coerces people into self-surveillance. There are existing unconventional ways of self-tracking that we can take inspiration from in order to inform future work, particularly in the realm of learning.

4.1 Unconventional and Qualitative Self-Tracking Alternatives

Let the People Label. Sharon and Zandbergen [25] discuss a self-tracking project called “Leaning into Grief” by Dana Greenfield. Dana created her own labels to capture the unique manifestations of her grief—she tracked the emotions that emerged in association with events, conversations, and sights that called to mind the memory of her late mother. Her methods made the act of self-tracking a personal, meaningful, and therapeutic experience; it drew up a blueprint of her life without the presence of her mother while soothing the ache of registering her absence. Even though the insights gleaned from the data gathered may only be relevant to Dana, it is this feature that makes it the most useful to her.

Accounting for the diversity of the human experience is possible not just in the realm of emotions but in other forms of self-tracking, simply by way of considering how users might be able to customize and humanize their experience through self-created labels and categories. The existence of pre-established labels is an obvious design decision: It helps us standardize analyses and make better comparisons between users. However, incorporating some forms of customization may help transform self-tracking methods into something people can make relevant to their lives. Rather than adjust behaviors to fit the system’s calculations, users can adjust the system to fit better with the parts of life they want to measure; they can also decide how those measurements are defined.

Autobiographical Tracking. Self-tracking can be beneficial without having an explicit purpose. For some, it may serve as a way to collect information purely for the sake of it. There is no act of reviewing the information afterwards, only the lingering comfort of knowing that it exists and has cataloged your life as things happened to you. For Alberto, the act of using outdated self-tracking methods and technology, as well as the decision to track something as innocuous as objects held in his right hand, allows him to resist the passivity that self-tracking pushes onto users and lets him swim against the tide of influence orchestrated by private corporations [25]. Tracking completely for oneself in a manner that tells a story can address the depersonalization that happens when a user is rendered a spectator in their data collection.

Context-Aware Computations. Self-tracking technologies could benefit from acknowledging factors beyond the control of the user and should account for the diversity of user circumstances. For example, a pregnant individual and an overweight person may have similar body compositions but will require different recommendations from their tracking device [3]; the technology should also be able to manage its expectations depending on the users’ condition, preferences, or special needs. Facets such as physical, mental, and psychological disabilities, preference for certain mindfulness and

fitness activities, and the user's level of commitment should be included in a device's consideration.

4.2 Flexible Tracking for Flexible Learning

Freeform Goal-Setting. Users may be more motivated to learn if they are allowed to incorporate their interests into the process of their study. They may also feel less pressured to conform to standardized behaviors if they can decide for themselves what their limits and expectations are; they can even work on eliminating the intimidating quality of learning something new by starting with learning something "useless." Simply contributing to a personal repository of interesting information with no technical or academic purpose, for example, can be a great way to encourage a love for gaining new knowledge. It also extinguishes the harmful idea that learning must only be purposeful. Whether the purpose is memorization, fast-paced learning, increased comprehension, or sustained productivity, allowing the user to decide the extent of their goals and how they wish to pursue them makes the process more attuned to their needs and specifications.

Learners Evaluate Themselves. Rather than provide learners with concrete and structured evaluation scores and push them to compare themselves with peers or classmates, learners can instead be asked to assess their own progress through open-ended questions reviewing the outcomes of past work. This allows them to reflect meaningfully on how far they have come and arrive at their own conclusions regarding any lingering points for improvement. Part of the issue of highly-quantitative self-tracking is that all the computations happen behind the screen—to let users become more active participants in their data collection, the data itself should be easy to peruse and understand, and options should exist that allow users to decide for themselves how well they are doing.

Scalability and the Option to Quit. Missing out on hours or days of tracking can very reasonably impact how well a tool understands its user. Such a instance, however, should not be punishing on the user in question—systems should be able to accommodate for absences or breaks without pressuring the user to change their behavior. On the contrary, it may be better if self-tracking tools had the ability to account for the varied contexts that may inevitably make learning difficult for a person: This includes their ability to show up consistently, to do certain things for a sustained period of time, or to handle the psychological load of mentally-challenging tasks. Difficulties and measurements should be scaled dependent on these factors, and reassurances should be given in the event that a user cannot satisfy even their own expectations. In the event that a user wants to opt out of interacting with a device, quitting should not feel like a moral failure—the technology simply was not for them, or they were simply not ready to integrate the technology into their lives.

ACKNOWLEDGMENTS

This work was supported by JST Trilateral AI Research (JP-MJCR20G3).

REFERENCES

- [1] Nike Ayo. 2012. Understanding health promotion in a neoliberal climate and the making of health conscious citizens. *Critical Public Health* 22, 1 (March 2012), 99–105. <https://doi.org/10.1080/09581596.2010.520692>
- [2] D. A. Baker. 2020. Four Ironies of Self-quantification: Wearable Technologies and the Quantified Self. *Science and Engineering Ethics* 26, 3 (June 2020), 1477–1498. <https://doi.org/10.1007/s11948-020-00181-w>
- [3] Matthew J. Bietz, Gillian R. Hayes, Margaret E. Morris, Heather Patterson, and Luke Stark. 2016. Creating Meaning in a World of Quantified Selves. *IEEE Pervasive Computing* 15, 2 (2016), 82–85. <https://doi.org/10.1109/MPRV.2016.39>
- [4] Christina E. Buse. 2010. E-scaping the ageing body? Computer technologies and embodiment in later life. *Ageing and Society* 30, 6 (Aug. 2010), 987–1009. <https://doi.org/10.1017/S0144686X10000164>
- [5] Fergal Carton, Patrick Brezillon, and Joseph Feller. 2016. Digital selves and decision-making contexts: towards a research agenda. *Journal of Decision Systems* 25, sup1 (June 2016), 96–105. <https://doi.org/10.1080/12460125.2016.1187416>
- [6] Eun Kyoung Choe, Nicole B. Lee, Bongshin Lee, Wanda Pratt, and Julie A. Kientz. 2014. Understanding quantified-selves' practices in collecting and exploring personal data. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM, Toronto Ontario Canada, 1143–1152. <https://doi.org/10.1145/2556288.2557372>
- [7] Brian G. Danaher, Håvar Brendryen, John R. Seeley, Milagra S. Tyler, and Tim Woolley. 2015. From black box to toolbox: Outlining device functionality, engagement activities, and the pervasive information architecture of mHealth interventions. *Internet Interventions* 2, 1 (March 2015), 91–101. <https://doi.org/10.1016/j.invent.2015.01.002>
- [8] Jean-François De Moya and Jessie Pallud. 2020. From panopticon to heautopticon: A new form of surveillance introduced by quantified-self practices. *Information Systems Journal* 30, 6 (Nov. 2020), 940–976. <https://doi.org/10.1111/isj.12284>
- [9] Annaleise Depper and P. David Howe. 2017. Are we fit yet? English adolescent girls' experiences of health and fitness apps. *Health Sociology Review* 26, 1 (Jan. 2017), 98–112. <https://doi.org/10.1080/14461242.2016.1196599>
- [10] Anthony Elliott. 2013. The Theory of New Individualism. In *Subjectivity in the Twenty-First Century* (1 ed.), Romin W. Tafarodi (Ed.). Cambridge University Press, 190–209. <https://doi.org/10.1017/CBO9781139035217.012>
- [11] Bruce Feiler. 2014. The United States of Metrics. *The New York Times* (May 2014). <https://www.nytimes.com/2014/05/18/fashion/the-united-states-of-metrics.html>
- [12] Peter E S Freund. 2004. Civilised Bodies Redux: Seams in the Cyborg. *Social Theory & Health* 2, 3 (Aug. 2004), 273–289. <https://doi.org/10.1057/palgrave.sth.8700031>
- [13] Sirkka L. Jarvenpaa and Karl R. Lang. 2005. Managing the Paradoxes of Mobile Technology. *Information Systems Management* 22, 4 (Sept. 2005), 7–23. <https://doi.org/10.1201/1078.10580530/45520.22.4.20050901/90026.2>
- [14] Luke Jones, Phil Marshall, and Jim Denison. 2016. Health and well-being implications surrounding the use of wearable GPS devices in professional rugby league: A Foucauldian disciplinary analysis of the normalised use of a common surveillance aid. *Performance Enhancement & Health* 5, 2 (Dec. 2016), 38–46. <https://doi.org/10.1016/j.peh.2016.09.001>
- [15] Evangelos Karapanos, Rúben Gouveia, Marc Hassenzahl, and Jodi Forlizzi. 2016. Wellbeing in the Making: Peoples' Experiences with Wearable Activity Trackers. *Psychology of Well-Being* 6, 1 (Dec. 2016), 4. <https://doi.org/10.1186/s13612-016-0042-6>
- [16] Tuomas Kari, Sanna Koivunen, Lauri Frank, Markus Makkonen, and Panu Moilanen. 2016. Critical Experiences During the Implementation of a Self-tracking Technology. Chiayi, Taiwan.
- [17] Ian Li, Anind Dey, and Jodi Forlizzi. 2010. A stage-based model of personal informatics systems. In *Proceedings of the 28th international conference on Human factors in computing systems - CHI '10*. ACM Press, Atlanta, Georgia, USA, 557. <https://doi.org/10.1145/1753326.1753409>
- [18] Steve Lohr. 2014. Unblinking Eyes Track Employees. *The New York Times* (June 2014). <https://www.nytimes.com/2014/06/22/technology/workplace-surveillance-sees-good-and-bad.html>
- [19] Deborah Lupton. 2012. M-health and health promotion: The digital cyborg and surveillance society. *Social Theory & Health* 10, 3 (Aug. 2012), 229–244. <https://doi.org/10.1057/sth.2012.6>
- [20] Deborah Lupton. 2013. *The Social Worlds of the Unborn*. Palgrave Macmillan UK, London. <https://doi.org/10.1057/9781137310729>
- [21] Dawn Nafus. 2014. This One Does Not Go Up to 11: The Quantified Self Movement as an Alternative Big Data Practice. *International Journal of Communication* 8 (2014), 1784–1794. <https://ijoc.org/index.php/ijoc/article/view/2170>
- [22] Parmy Olson. 2014. Wearable Tech Is Plugging Into Health Insurance. <https://www.forbes.com/sites/parmyolson/2014/06/19/wearable-tech-health-insurance.html>
- [23] Andrea Parrish. 2012. Gamification Keeps Me Going (aka I'm the Self-Tracking Type): An Andrea Post. <http://bikestylespokane.com/2012/06/09/gamification-keeps-me-going-aka-im-the-self-tracking-type-an-andrea-post/>
- [24] Minna Ruckenstein. 2014. Visualized and Interacted Life: Personal Analytics and Engagements with Data Doubles. *Societies* 4, 1 (Feb. 2014), 68–84. <https://doi.org/10.3390/soc4010068>
- [25] Tamar Sharon and Dorien Zandbergen. 2017. From data fetishism to quantifying selves: Self-tracking practices and the other values of data. *New Media & Society* 19, 11 (Nov. 2017), 1695–1709. <https://doi.org/10.1177/1461444816636090>

- [26] Siew Kien Sia, May Tang, Christina Soh, and Wai Fong Boh. 2002. Enterprise resource planning (ERP) systems as a technology of power: empowerment or panoptic control? *ACM SIGMIS Database: the DATABASE for Advances in Information Systems* 33, 1 (Feb. 2002), 23–37. <https://doi.org/10.1145/504350.504356>
- [27] Per-Anders Tengland. 2016. Behavior Change or Empowerment: On the Ethics of Health-Promotion Goals. *Health Care Analysis* 24, 1 (March 2016), 24–46. <https://doi.org/10.1007/s10728-013-0265-0>
- [28] Kaiton Williams. 2013. The Weight of Things Lost: Self-knowledge and Personal Informatics. *Personal Informatics* (April 2013), 4.